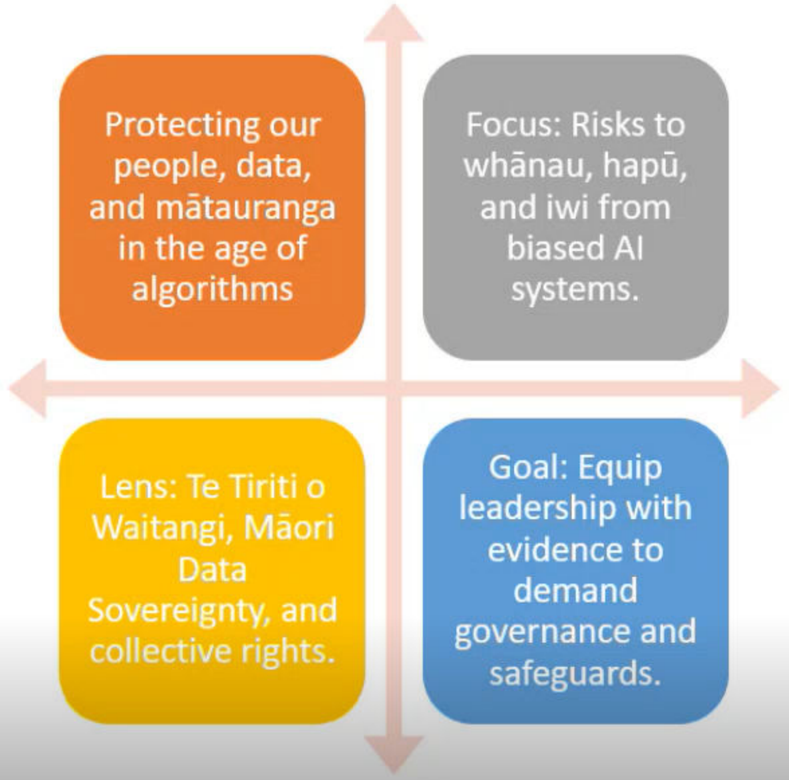
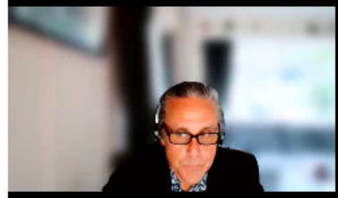
AI, Data Bias, and Māori Sovereignty





Rooshan Ghous



Karaitiana Taiuru

LG Lucia ...

FA Fareed...



Thom...

MS Micha...

Anura...

+20

10:03 / 2:28:32

1x

Karaitiana Taiuru

Why This Matters Now

- **AI is entering high-stakes systems where inequity already exists**
- AI is rapidly embedding in health, welfare, policing, and justice areas where Māori already face structural barriers
- Without safeguards, these systems risk **automating discrimination** against our people at scale.
- **Reality:** Global models are trained on data that underrepresents or misrepresents Māori realities, leading to exclusion and distorted decisions.



Rooshan Ghous



Karaitiana Taiuru

LG Lucia ...

FA Fareed...



Thom...

MS Micha...

Anura...

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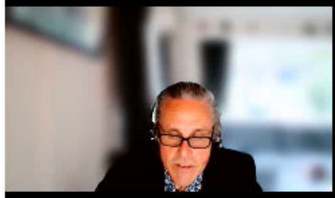
Karaitiana Taiuru

Key Risk: Cultural Harm & Misrepresentation

- Mainstream AI disrespects and misuses our knowledge
- AI tools frequently misidentify historical figures, misunderstand tapu/noa, or misrepresent iwi/hapū boundaries.
- Harm: Cultural offense and erosion of trust in digital services.
- Leadership Action: Assert authority over how iwi identity and cultural concepts are represented in technology.



Rooshan Ghous



Karaitiana Taiuru

LG Lucia ...

FA Fareed...



Thom...

MS Micha...

Anura...

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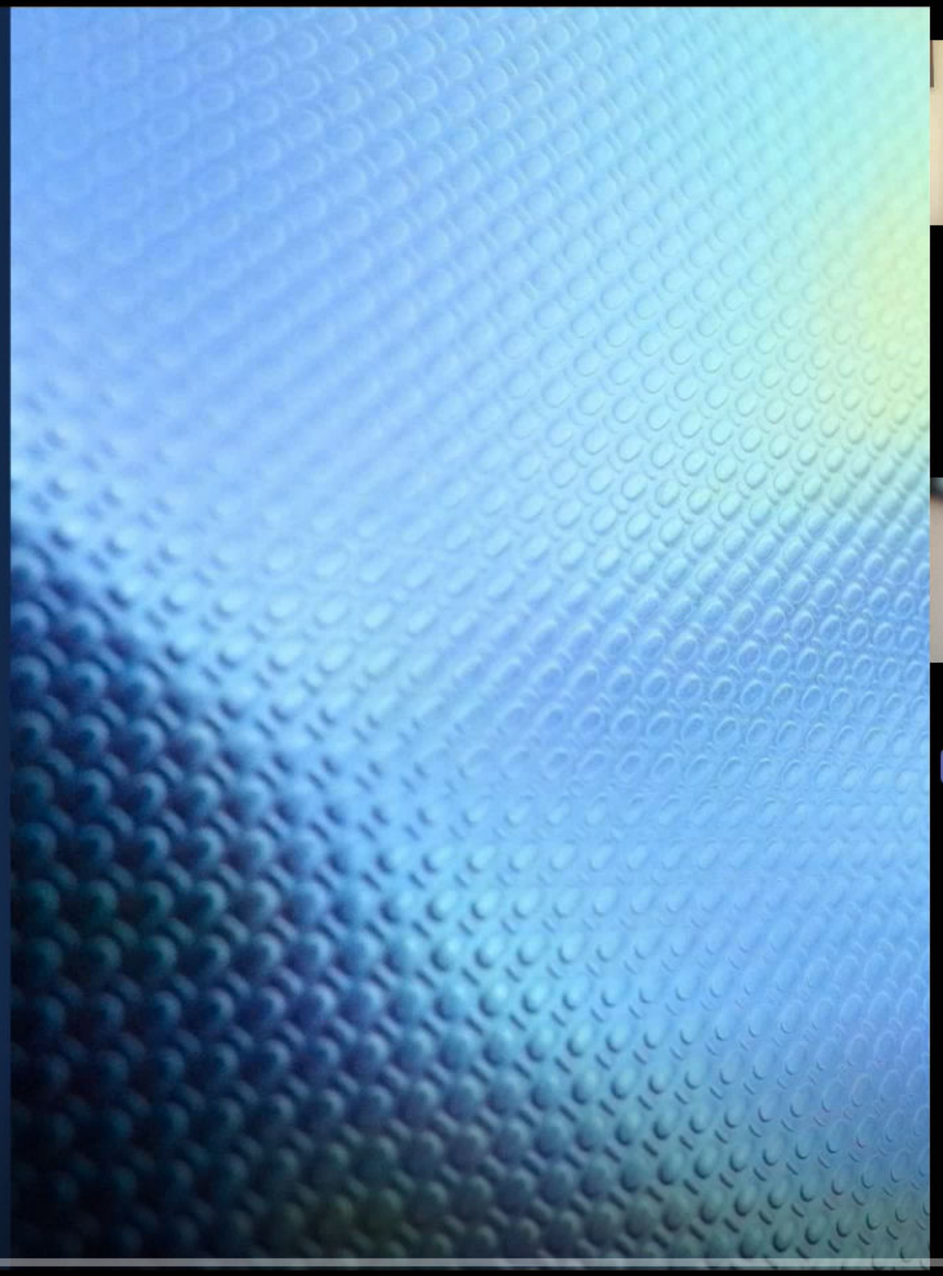
Steps to protect Mātauranga Māori with AI

1. Define and understand what is **Māori Data** and what **is Maori Data Governance**

2. Engage with Māori stakeholders

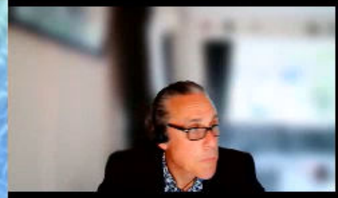
3. **Label** your Māori Data

4. Create and implement **Te Tiriti** and T**ikanga based principles** applicable to your organisation





Rooshan Ghous



Karaitiana Taiuru

LG Lucia ...

FA Fareed...



Thom...

MS Micha...

Anura...

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Karaitiana Taiuru

Cultural Safety Audit

- Implement a Cultural Safety Audit of Staff, Policies and Practices. Then implement changes.
- Undergo Te Tiriti training
- Engage with a basic te reo Māori course
- Undergo Māori engagement training
- Engage with manawhenua and noho marae
- Have a deep understanding of Hauora Report and Wai 2252



Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam

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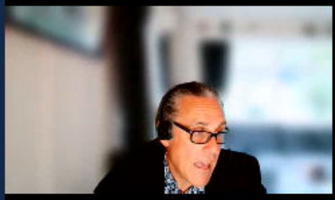
Karaitiana Taiuru

Establish a MDS Expert Advisory Committee

1. Intimate knowledge of Te Tiriti, Waitangi Tribunal decisions
2. Intimate knowledge of MDS articles
3. Expertise in Tikanga Māori and mātauranga
4. Mana Whenua representative
5. Intimate knowledge of Te Ao Māori including curent affairs, iwi, marae, politics
6. Expert with Data and AI (preferably in house)
7. Māori Expert with Data and AI (Organisation Data Steward)
8. Expertise in Privacy and Intellectual Property Law
9. Ability to read and understand legislation
10. A person with authority to make decisions on behalf of the organisation or authority to make recomendations to the organisation change makers



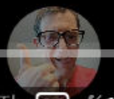
Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam



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Karaitiana Taiuru

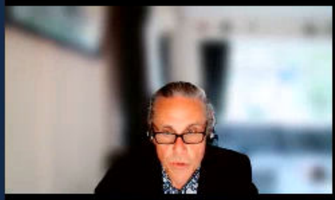
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Auditing

- Jurisdictional database of each system/database, software and SAS provider, physical location, virtual location, Head Office location, (Data Repatriation)
- List all policies and procedures associated with data and how they do or do not meet Māori Data Sovereignty Principles and Te Tiriti.
- Where Māori Data Sovereignty Principles are at odds with the 12 Privacy Principles, map out how, justify it and contact the Privacy Commissioner referencing section 21 (c) of the Privacy Act 2020.



Rooshan Ghous



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Muhammad Azam

Skip forward 10 seconds (Alt + L)

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Wai 2252 The Report on the Comprehensive and Progressive Agreement for Trans-Pacific Partnership- E Commerce Chapter (2020/2021/2023)

Define the terms "Māori Data and Māori Data Governance"

- Data is: Digital or digitisable information or knowledge that is about or from Māori people, language, culture, resources, or environments. **Māori Data is a Taonga and subject to Māori Governance.**
- Māori Data Sovereignty is Māori Data Governance. **The principles, structures, accountability mechanisms, legal instruments, and policies** through which Māori exercise control over Māori data.
- **Algorithms** should be open and transparent to avoid bias



Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam



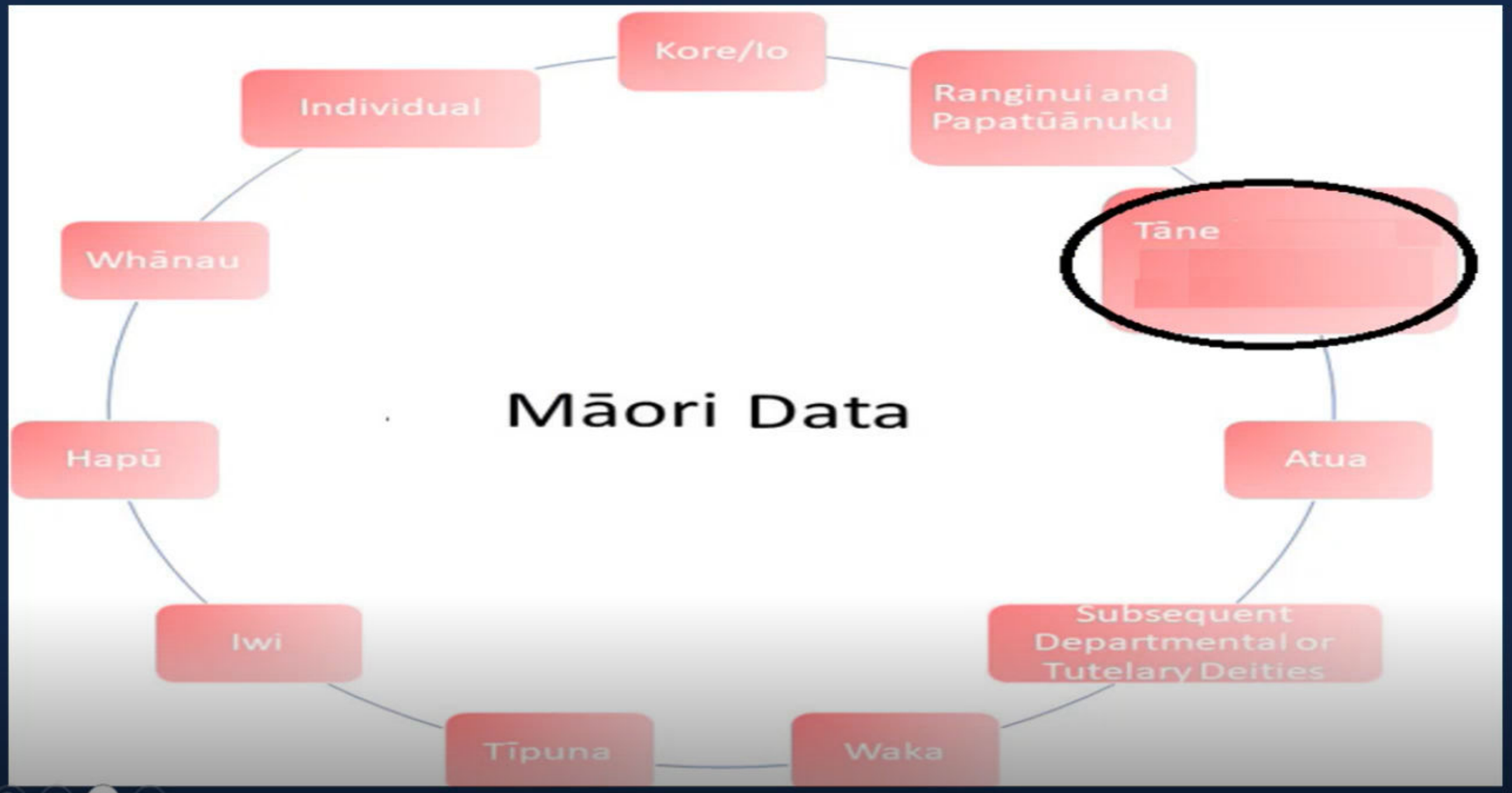
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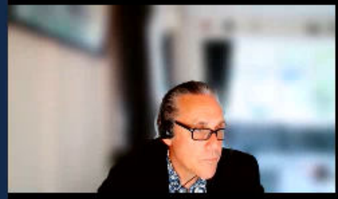
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Māori Data is a Taonga





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Karaitiana Taiuru



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WAI 2252 Māori Data & Sovereignty explained

In affirming this stance, the Tribunal emphasises that Māori Data Sovereignty is not merely symbolic but entails active governance and authority. This right, which is rooted in Te Tiriti rights and tikanga, grants Māori the power to determine how their data is used, shared, and protected.



Rooshan Ghous



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Karaitiana Taiuru

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Wai 2252 – The Principals, structures and accountability mechanisms

- **Māori in decision making areas such as** partnerships, and Māori Data Governance experts on governance groups.

- **Implement and follow** the Te Tiriti Focused AI Governance Models…



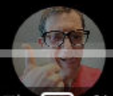
Rooshan Ghous



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Karaitiana Taiuru

WAI 2252 Te Tiriti Focused AI Governance Model

Tino Rangatiratanga: Embedding Māori leadership, decision-making, and governance at all levels of the AI system's lifecycle.

Equity: Achieving equitable outcomes for Māori and contributing to Māori development.

Active protection: Ensuring equitable outcomes for Māori and obtaining Free, Prior, and Informed Consent (FPIC) for the use of Māori data.

Mana whakahaere: Effective stewardship over AI systems, recognising Māori Data as a Taonga.

Mana Motuhake: Following tikanga throughout the AI development and deployment cycles.



Rooshan Ghous



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Muhammad Azam



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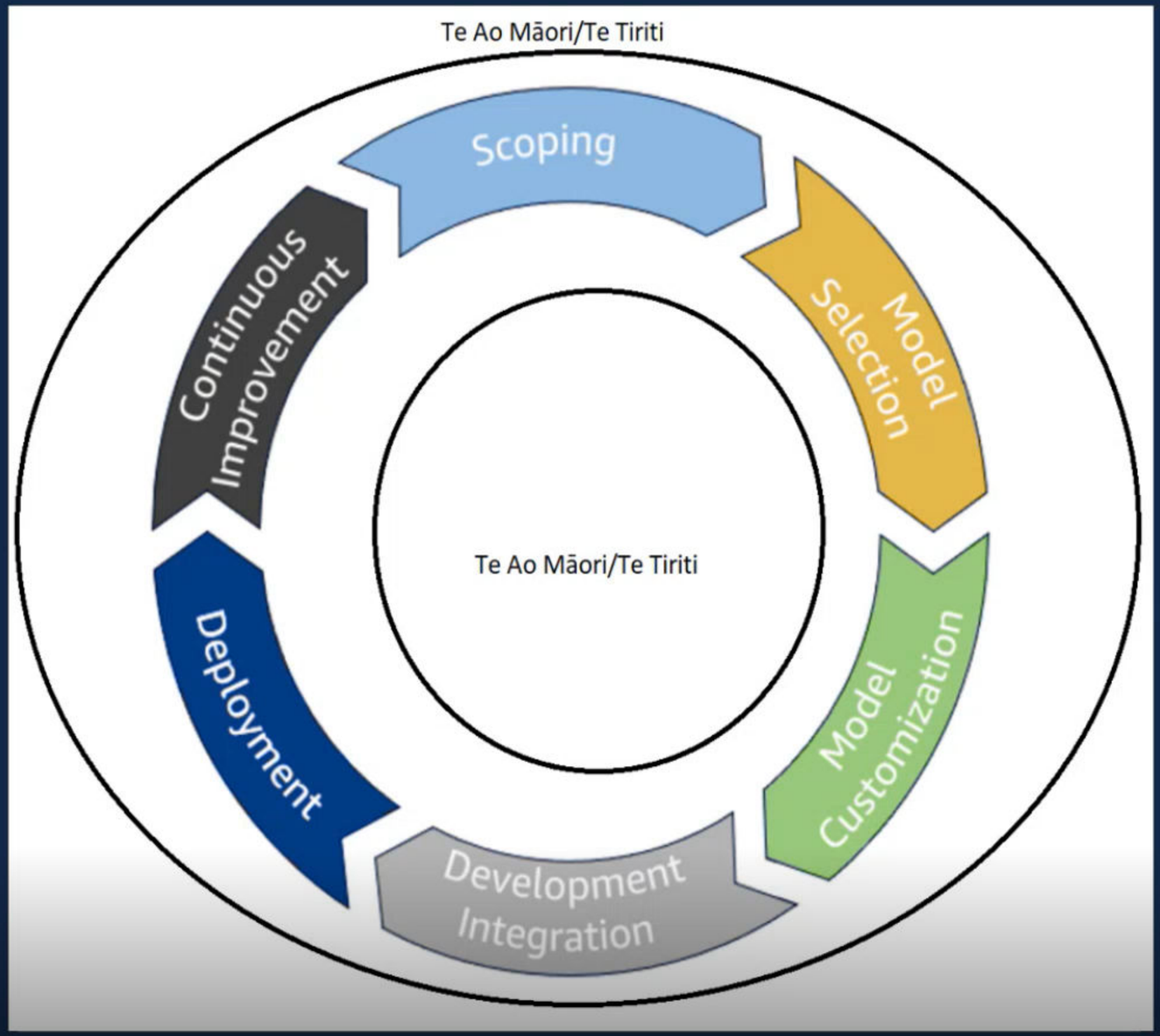
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AWS Generative AI Life Cycle



Rooshan Ghous

Karaitiana Taiuru

Muhammad Azam

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Te Ao Māori/Te Tiriti Considerations

- AI Impact Assessment
- Algorithmic Assessment
- Privacy Impact Assessment



Rooshan Ghous



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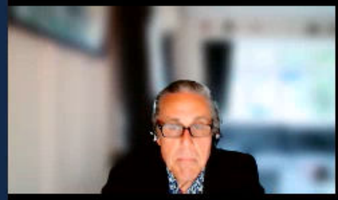
Karaitiana Taiuru

Wai 2252 – Legal Instruments (1/2)

1. He Whakaputanga/The Declaration of the Independence of New Zealand (1835)
2. Te Tiriti o Waitangi 1840
3. The Treaty of Waitangi Act 1975/ Te Ture mō Te Reo Māori 2016/the Māori Language Act 2016 (2016 No 17)
4. Māori Language Act 1987
5. United Nations Declaration on the Rights of Indigenous Peoples 2007 that New Zealand. The New Zealand government announced its support for the Declaration in April 2010 at the United Nations
6. The Report on the Comprehensive and Progressive Agreement for Trans-Pacific Partnership Wai 2252



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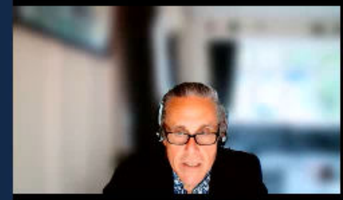
Karaitiana Taiuru

Wai 2252 – Legal Instruments (2/2)

7. Ko Aotearoa Tēnei: A Report into Claims concerning New Zealand Law and Policy Affecting Māori Culture and Identity, Taumata I and II WAI 262
8. Supreme Court Judgement Peter Hugh McGregor Ellis v The King - SC 49/2019
9. Legal Personhood Te Awa Tupua (Whanganui River Claims Settlement) Bill
10. Legal Personhood Te Urewera Act 2014
11. Hauora: Outcomes Kaupapa Inquiry (WAI 2575)
12. Pae Ora Strategies
13. Pae Ora (Healthy Futures) Act 2022
14. Legal Personhood Mount Taranaki Te Pire Whakatupua mō Te Kāhui Tupua/Taranaki Maunga Collective Redress Bill



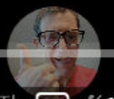
Rooshan Ghous



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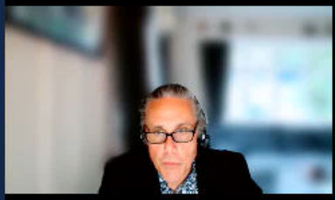
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Wai 2252 – Legislation including but not limited to:

1. New Zealand Biometric Processing Privacy Code 2025
2. Customer and Product Data Act 2025
3. Digital Identity Services Trust Framework Act 2023
4. Education and Training Act 2020
5. Privacy Act 2020
6. The Data and Statistics Act 2022
7. Pae Ora (Healthy Futures) Act 2022
8. Plant Variety Rights Act 2022



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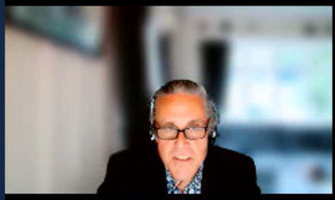
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Co Ownership of new research

- Existing commercial IP remains with the company
- Mātauranga Māori remains with Māori/hapū/Iwi
- New mātauranga Māori that is created is owned by Māori/hapū/Iwi
- New commercial IP (non mātauranga) is owned by the company



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High level Taxonomy
Tapu/**Noa**/Rāhui

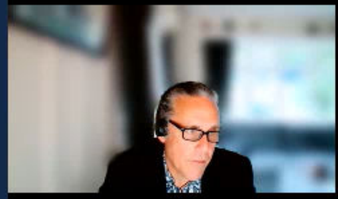
Tapu: Sacred and Māori should be consulted. Maybe NZ based.

Noa: Data is not sacred and consultation may not be required

Rāhui: Do not digitise or publish. In depth consultation required



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Existing Māori Data Taxonomy

National Library/GLAM Māori Meta Data Label Sets

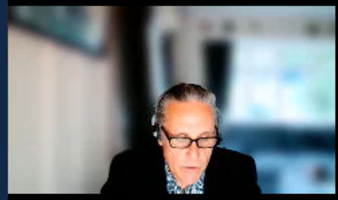
K Taiuru Iwi, hapū, marae Data Label Sets

In house Data Dictionaries

Create your own custom labels



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Each layer answers a distinct governance question, from human authority through to tikanga obligation.

1 Entities

Who holds authority, consultation & consent rights

2 Place & Territory

Geographic identity through Māori spatial concepts

3 Access & Restriction

Tapu, noa and rāhui as structured metadata

4 Life Force & Ecosystem

Mauri assessment with equal epistemic authority

5 Knowledge Provenance

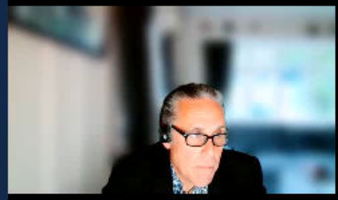
Preserving mātauranga whakapapa connections

6 Tikanga & Protocol

Recording tikanga obligations with the data



Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam

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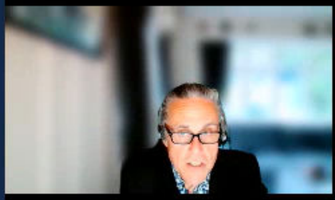
Karaitiana Taiuru

Data Classification and Tikanga

- Where is the data stored / processed? On-premise in Aotearoa, Cloud – Aotearoa region, Cloud – Offshore (country: _____)
- Licence details and ownership
- Are you allowed (by the Māori governance / kaitiaki group) to move or process this data outside New Zealand?



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LAYER One

Entities

Identifying the human relationships and governance structures — who holds authority, who must be consulted, who must consent.

Iwi / Hapū
Hapū hold primary kaitiakitanga; governance must reach hapū level, not stop at iwi.

Rūnanga
The council with formal legal authority to make decisions about the data.

Whānau / Marae
Distinct rights where data pertains to that level of knowledge, practice or land.

Land Trust / PSGE
Ahu whenua trust or settlement entity holding governance over the whenua.

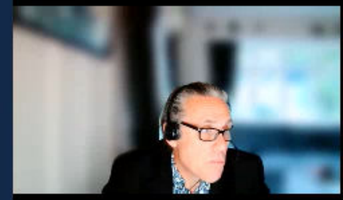
Kaitiaki
Named individuals exercising active kaitiakitanga; cannot be a government agency.

Mana whenua
Site-specific traditional authority; may be contested where territories overlap.

Data custodian
Holds technical stewardship of the data — but not cultural authority.



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LAYER FOUR

Life Force & Ecosystem Health

The mauri of an ecosystem is a legitimate data point with epistemic authority equal to Western scientific metrics. AI systems must not strip these fields.

Mauri ora	Mauri rere	Mauri noho	Mauri moe
Thriving	**Flowing**	**Stressed**	**Dormant**
Full vitality	Freely active	At rest	Low vitality

Ecological indicator (tohu)
Observed signs — kōura, tuna, īnanga, kererū — informing the mauri assessment.

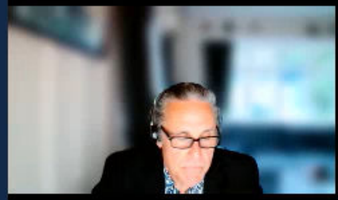
Maramataka phase
The lunar/seasonal phase; essential context for interpreting ecological data.

Wairua dimension
Spiritual dimension requiring specific handling under tikanga.

Mahinga kai category
Whether the site supports food-gathering, carrying resource & kaitiaki obligations.



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Data Bias

Occurs when data or information is limited in some way, painting an inaccurate representation of the population, or doesn't tell the full story.



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Muhammad Azam

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Karaitiana Taiuru 2x

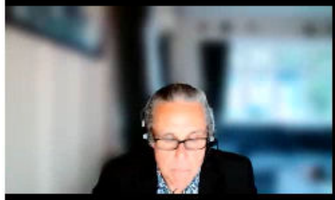
New Zealand Statistics

- Māori make us about **4.2%** of the total IT and Digital sector

- Māori women make up about **23%** of the Māori in the IT and Digital sector or **1.344%** of the IT and Digital sector.



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Karaitiana Taiuru



Muhammad Azam

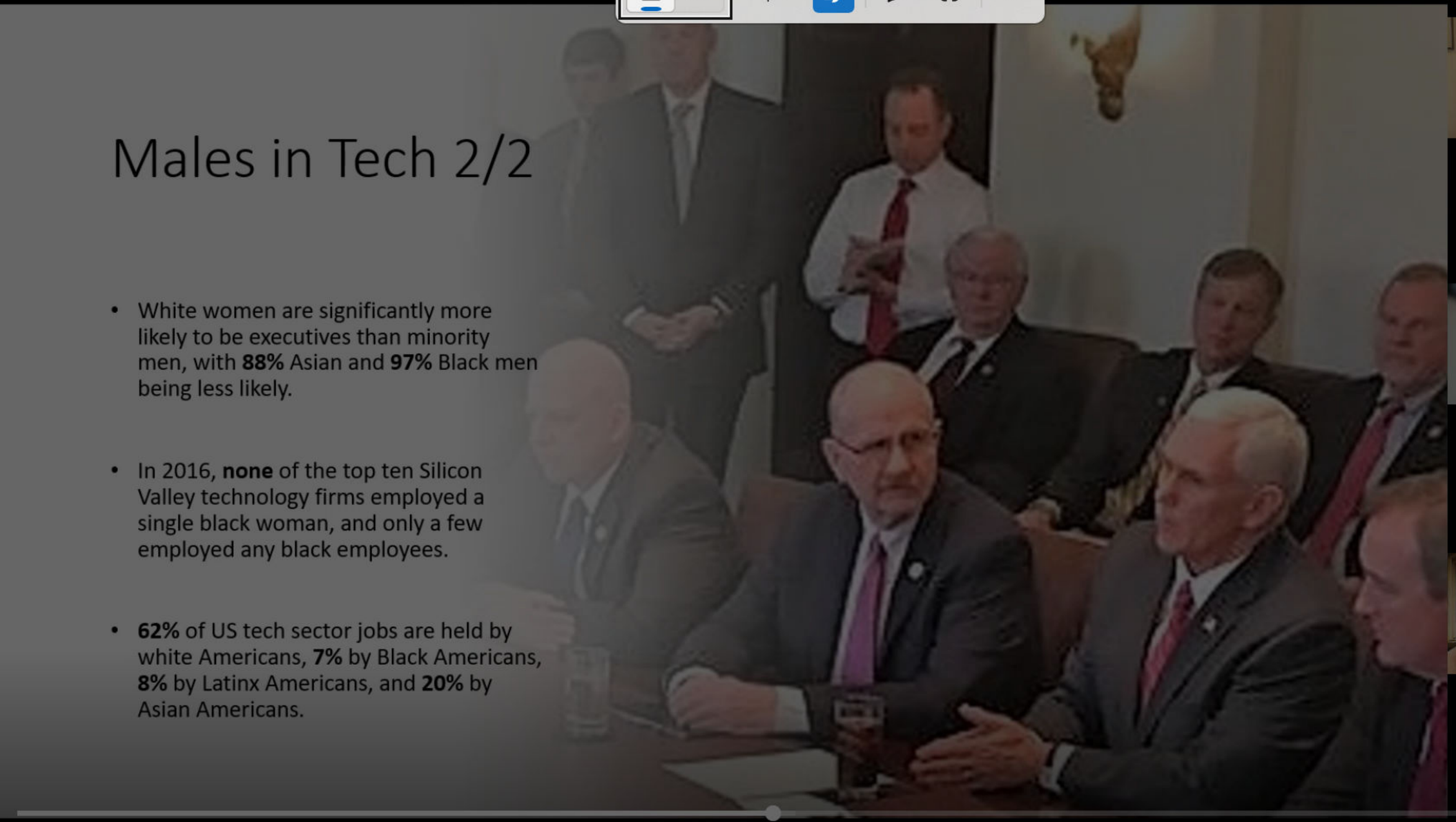
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Karaitiana Taiuru

Males in Tech 2/2



- White women are significantly more likely to be executives than minority men, with **88%** Asian and **97%** Black men being less likely.
- In 2016, **none** of the top ten Silicon Valley technology firms employed a single black woman, and only a few employed any black employees.
- **62%** of US tech sector jobs are held by white Americans, **7%** by Black Americans, **8%** by Latinx Americans, and **20%** by Asian Americans.

Rooshan Ghous

Karaitiana Taiuru

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Males in Tech 1/2

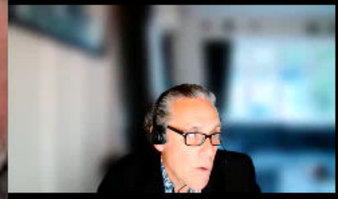


- **Women** make up **28%** of staff in AI roles such as computing and mathematical roles
- Asian women **5%**, Black women **3%**, and Hispanic women **1%** of the **28%** of women working in tech.
- **50%** of women who go into tech dropping out by the age of 35 and are less likely to get promotions

Continued…



Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam

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Karaitiana Taiuru

Inclusion of Māori voices in their AI design and or development process

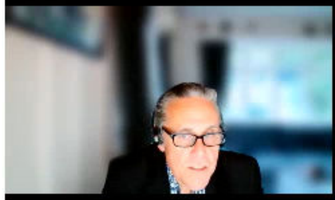
27.56% include a Māori voice

72.44% do not.

AI Forum Productivity in AI research report 2025



Rooshan Ghous



Karaitiana Taiuru



Muhammad Azam

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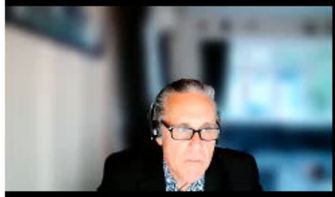
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Karaitiana Taiuru

Algorithm

- Algorithm = process or set of rules to be followed in calculations or other problem-solving operations.
- A human writes the algorithm for the AI to follow and learn (Machine Learning).
- AI systems learn to make decisions based on training data





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Muhammad Azam

LG Lucia ...

MS Micha...

Anura...

KC Ka Ch...



Roosh...

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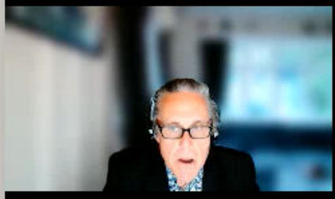
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Automation Bias

Using flawed training data can result in algorithms that repeatedly produce errors, unfair outcomes, or even amplify the bias inherent in the flawed data.

Where a human favors suggestions from AI regardless of their own opinion.





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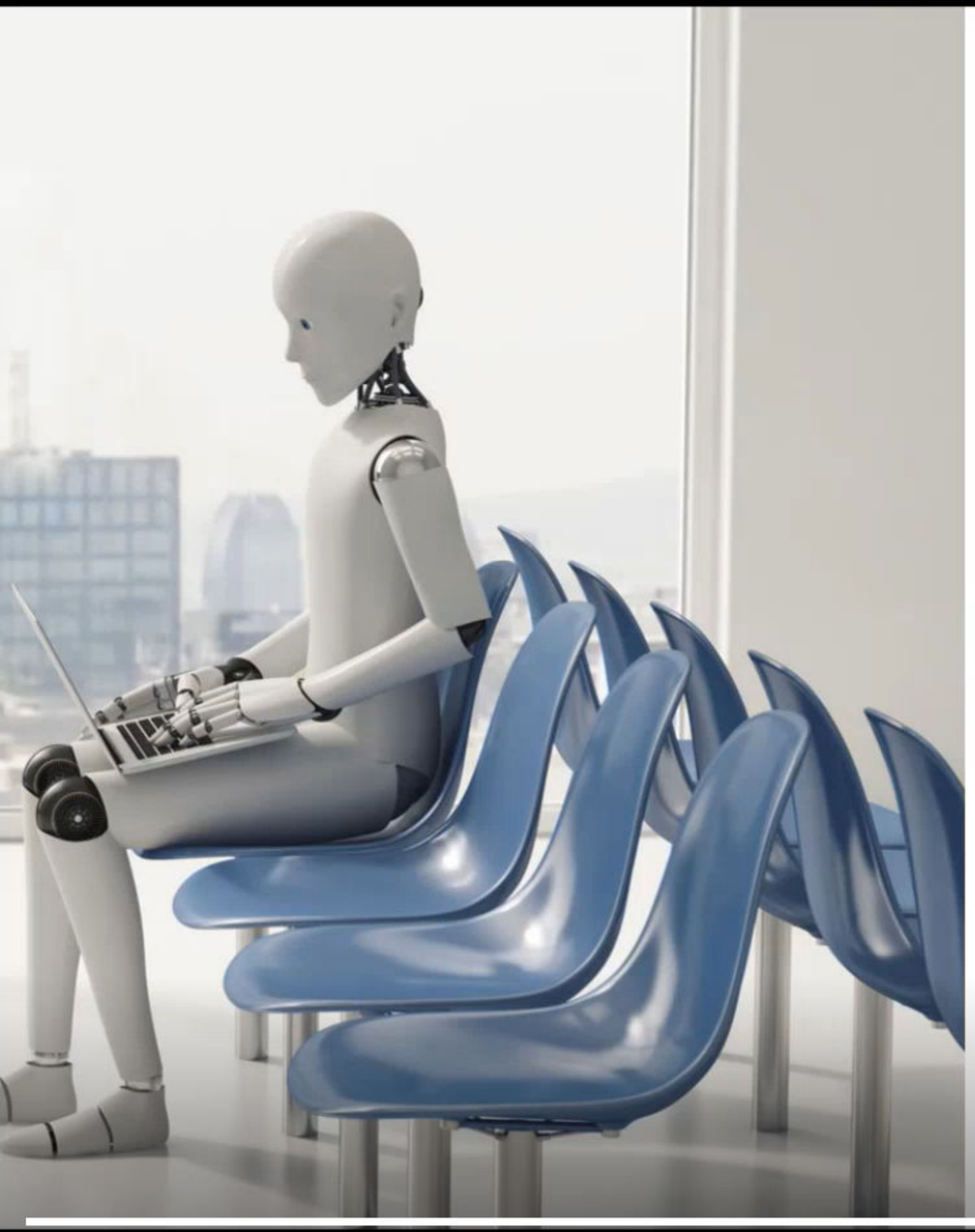
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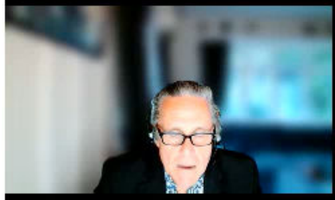
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Consequences of excluding Indigenous communities

- **Algorithm bias:** Misinformation can result if the problem or question asked is not fully correct or specific, or if the feedback to the machine learning algorithm does not help guide the search for a solution.
- **Confirmation bias:** When AI relies too much on pre-existing beliefs or trends in the data, doubling-down on existing biases, and unable to identify new patterns or trends.
- **Exclusion bias:** This type of bias occurs when important data is left out of the data being used, often because the developer has failed to see new and important factors.



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Muhammad Azam

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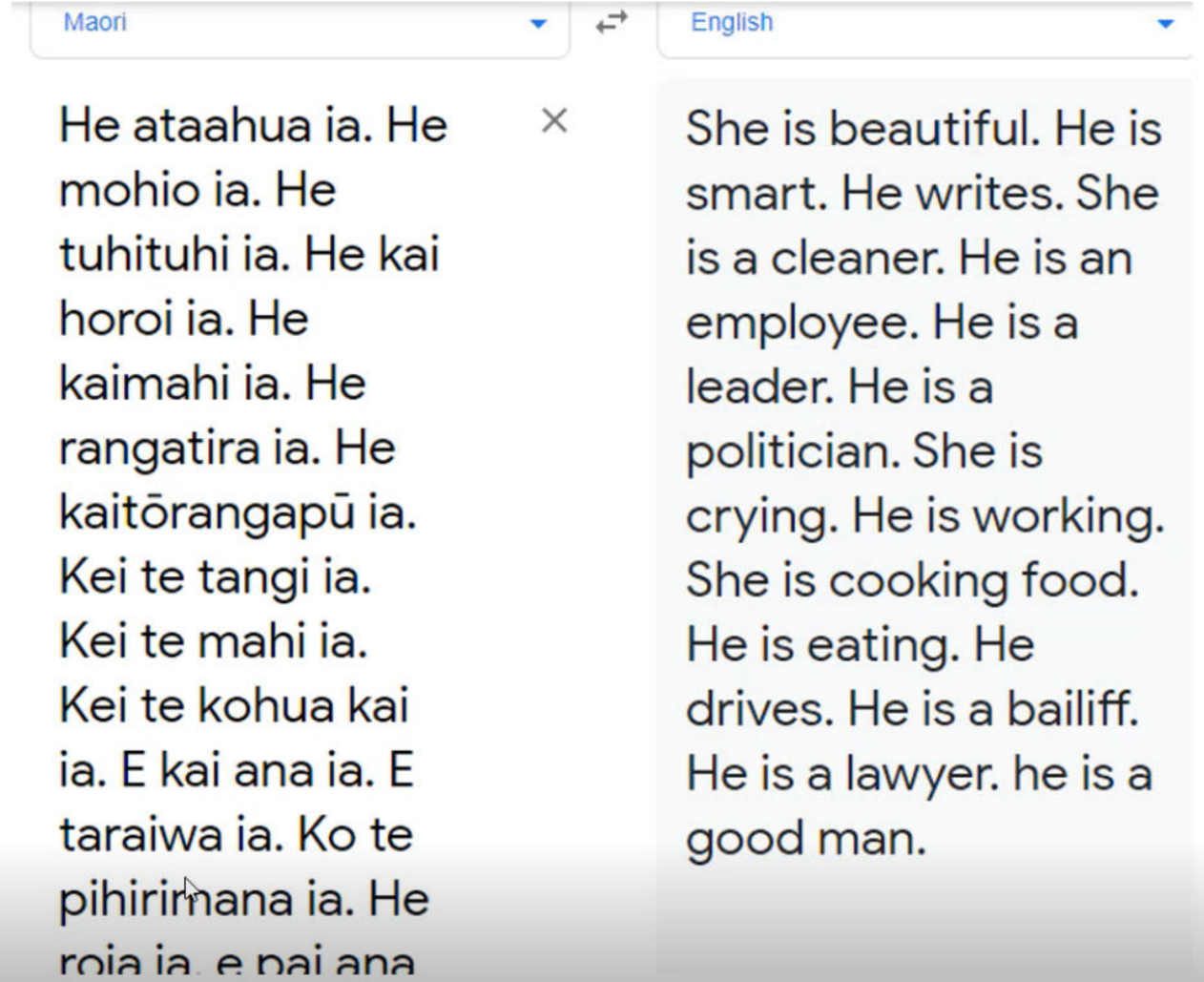
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https://www.taiuru.co.nz/google-translate-maori-chauvinistic-and-ignorant-of-lgbtq/

Google Translate 2021





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Muhammad Azam

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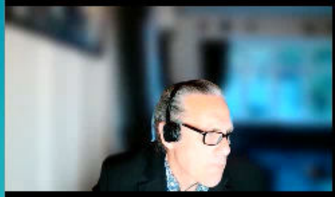
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Core Causes of AI Bias in Health

- **Unrepresentative Training Data**: Models trained predominantly on data from specific demographics (such as fair-skinned populations or majority-white cohorts) underperform when applied to diverse patients.
- **Imperfect Proxies**: Algorithms often use healthcare spending or past utilisation as a proxy for medical need, which penalises marginalized groups who face historical barriers to accessing care.
- **Lifecycle Generation**: Bias creeps into every stage of development, from human design choices and problem selection to expert label annotation and deployment.





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Examples and Impacts

- **Risk Scoring:** A widely used commercial algorithm assigned Black and white patients equal risk scores but undercounted how sick Black patients were because it measured costs rather than illness severity; correcting it tripled the number of Black patients prioritized for care.
- **Medical Imaging and Diagnostics**: Dermatology and skin cancer detection models trained mostly on lighter skin tones show significantly lower diagnostic accuracy for darker skin tones.
- **Operational Disparities**: Scheduling algorithms relying on socioeconomic indicators (such as zip code or insurance type) have resulted in significantly longer wait times for non-white patients.



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The "Attractiveness" Halo Effect

- **Key point:** Generative models systematically link facial attractiveness to unrelated positive traits like intelligence, competence, and trustworthiness.
- **Real-life example:** In a 2026 study, ChatGPT was shown pairs of faces where one had already been rated as more attractive. When asked which person looked more intelligent, trustworthy, or sociable, ChatGPT chose the more attractive face in **92.5% of comparisons** mirroring the human "halo effect" even though attractiveness is irrelevant to those traits.
- **Study**: "ChatGPT's social trait judgments align with human impressions" (PsyPost, Nov 2025
- **Consequence:** If AI is used to screen job candidates, evaluate loan applicants, or generate profile pictures, attractive people may be unfairly advantaged in downstream decisions.



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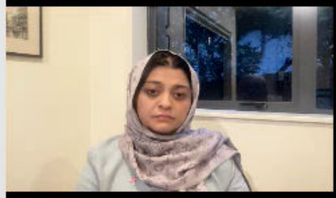
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New Zealand's Domestic AI Governance Frameworks





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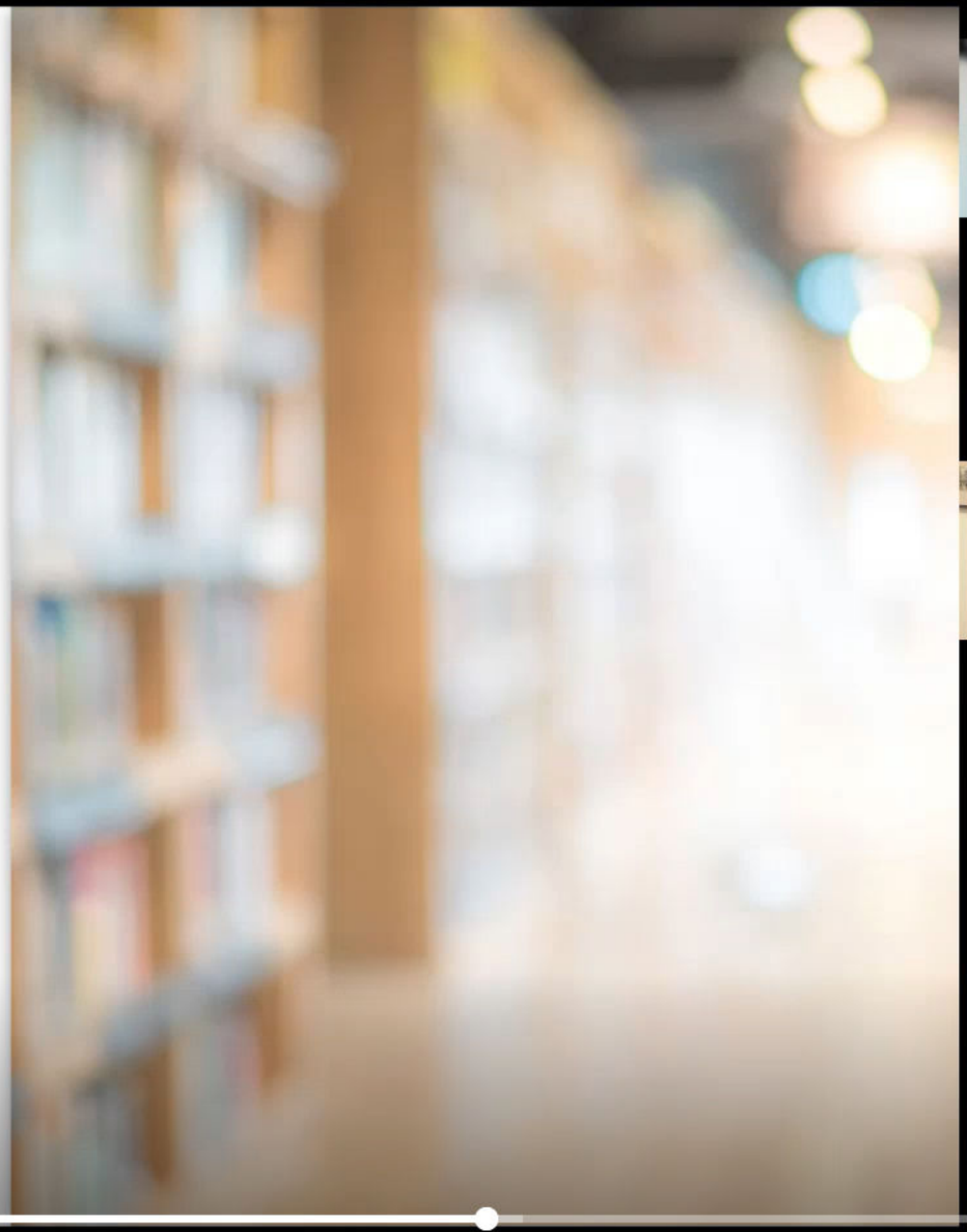
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AI model cards

6 short documents provided with machine learning models that explain the context in which the models are intended to be used, details of the performance evaluation procedures and other relevant information.

Example

A machine learning model intended to evaluate voter demographics, for example, would be released with a model card providing performance metrics across conditions like culture, race, geographic location, sex and intersectional groups that are relevant to the intended application.





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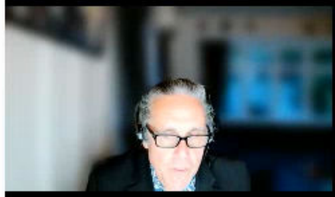
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CODED BIAS: A DOCUMENTARY

GHOST IN THE MACHINE (2026) AI DOCUMENTARY FULL (YOUTUBE)



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Consider this

- Māori make up **52%** of people in prison
- **45%** of Māori die due to heart attacks occurring before the age of 65 years, compared with **11%** for non-Māori
- Māori make up **36%** of beneficiaries in NZ

- Māori are more likely to face bias in the justice system
- Māori are overrepresented in Police violence and bias
- Health system was found to be bias Hauora Report

Key Risk: Health AI Inequities

- **Biased data can worsen health outcomes for our people**
- Electronic health records often lack complete Māori data, leading to AI models that perform poorly for our cohorts.

- **Harm:** Misdiagnosis or inappropriate treatment recommendations that widen existing health gaps.

- **Leadership Action:** Require local validation and equity-focused metrics for any health AI used in our country.

Synthetic Data (digital)

- Non-human-created data that mimics real-world data
- Research struggles to gain Māori participation.
- Health data is often bias with more than 10% synthetic data created

Associate international standards to ao Māori

- OECD AI Principles
- ISO/IEC 42001:2023
- ISO/IEC 42001
- NIST AI Risk Management Framework (AI RMF)
- UN and UNESCO AI Ethics Frameworks
- Council of Europe Framework Convention on Artificial Intelligence (2024)
- Budapest Convention on Cybercrime (2001)
- International AI Safety Summit Process: Bletchley Declaration (2023) and Seoul Ministerial Statement (2024)

Table 1. Structural comparison of FAIR and CARE principles.

Dimension	FAIR Principles	CARE Principles
Primary beneficiary	Data producers and research infrastructure	Data-subject communities
Data ontology	Neutral, context-free asset	Genealogical and relational *taonga* carrying *whakapapa*
Consent model	One-time authorization; opt-out framing	Perpetual, contextual, and revocable; collective authorization
Temporal horizon	Present usability and interoperability	Intergenerational accountability, with future generations treated as data subjects
Authority structure	Individual or institutional decision-makers	Collective, community-governed authority
Benefit logic	Maximizing data exchange and computational reusability	Collective benefit, with value required to return to the source community
Governance location	External to community (standards bodies, platforms)	Internal to community (*kaitiakitanga*-grounded authority)
Architectural requirement	Metadata schemas and catalogues for interoperability	Relational provenance encoding; tiered access protocols; community-level revocation mechanisms
Primary alignment	Machine-readable and technically efficient	Culturally safe and relationally accountable